

Prefill-Context Execution Eval

20-Task Report v3: Cleaned Data + Honest Analysis

Model: Claude Sonnet 4.6 | 2026-05-15 | SkillsBench Dataset

0. CRITICAL: Data Quality Issues

WARNING: 22 prefill configs and baseline configs were rate-limited (Max Plan daily limit hit). These trials returned rc=1 with 'hit your limit' in stdout and wall_time ~1s. They are EXCLUDED from all analysis below.

Affected tasks (prefill): azure-bgp-oscillation-route-leak (100% poisoned), citation-check (100%), civ6-adjacency-optimizer (100%), energy-market-pricing (75%), energy-ac-optimal-power-flow (67%), adaptive-cruise-control (10%).

Affected baselines: 2/14 noskill, 11/14 gtonly trials rate-limited. GT-only has only 3 valid results - NO CONCLUSIONS can be drawn from it.

Previous versions of this report incorrectly counted rate-limited trials as FAILs, producing fake '0% pass rate' for 3 tasks and a fake 'GT-only paradox' (14% pass rate). These findings are RETRACTED.

1. Experiment Overview

This experiment measures how Claude Sonnet 4.6 performs software engineering tasks when procedural SKILL.md documents are injected into the system prompt. Three conditions: Prefill (GT + noise skills), Noskill (no skills), GT-only (GT skills only).

After Cleaning

Prefill clean configs	146 (of 168 attempted, 22 rate-limited)
Noskill N=5 clean	13 tasks, 13 clean trials
GT-only N=5 clean	3 tasks, 3 trials (59/70 rate-limited - UNRELIABLE)
Tasks with full 12/12 prefill	8 of 14
Tasks with partial prefill	3 (energy-ac 4/12, energy-market 3/12, enterprise 11/12)

2. Prefill Results (Clean Data Only)

Pass Rate	96/146 = 65.8%
PASS	96
FAIL	50
Avg Wall Time	320s (median 187s)
PASS Avg Wall	273s
FAIL Avg Wall	411s

3. Per-Task Results (Clean)

Task	Cfgs	Pass	Rate	NS	GT	Wall
3d-scan-calc	12/12	12/12	100%	1/1	1/1	72s
earthquake-plate-calculation	12/12	12/12	100%	0/1	RL	117s
energy-market-pricing	5/12	5/5	100%	RL	RL	287s
dialogue-parser	12/12	11/12	92%	0/1	RL	244s
citation-check	12/12	11/12	92%	1/1	RL	154s
dapt-intrusion-detection	12/12	10/12	83%	0/1	RL	102s
adaptive-cruise-control	12/12	9/12	75%	0/1	0/1	756s
earthquake-phase-association	12/12	9/12	75%	1/1	RL	673s

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econ-detrending-correlation	12/12	9/12	75%	1/1	RL	117s
energy-ac-optimal-power-flow	5/12	2/5	40%	0/1	RL	303s
exceltable-in-ppt	12/12	4/12	33%	0/1	RL	202s
azure-bgp-oscillation-route-..	5/12	1/5	20%	0/1	0/1	503s
civ6-adjacency-optimizer	11/12	1/11	9%	0/1	RL	791s
enterprise-information-searc..	12/12	0/12	0%	0/1	RL	279s

NS=Noskill N=5 (pass/total), GT=GT-only N=5 (pass/total), RL=Rate-limited (no data). GT-only has only 3 tasks with clean data due to rate limiting - mostly unusable.

4. Preliminary Observations (NOT Causal Claims)

The following are observations from the CLEAN data only. They are NOT causal claims. All baseline comparisons have N=1 and are statistically meaningless for individual tasks. We report them only as directional signals requiring validation.

4.1 Tasks where prefill clearly helps (noskill<30%, prefill>=50%)

- adaptive-cruise-control: noskill=0/1 (0%), prefill=75% (9/12)
- dapt-intrusion-detection: noskill=0/1 (0%), prefill=83% (10/12)
- dialogue-parser: noskill=0/1 (0%), prefill=92% (11/12)
- earthquake-plate-calculation: noskill=0/1 (0%), prefill=100% (12/12)

4.2 Tasks where both succeed (noskill>=50%, prefill>=50%)

- 3d-scan-calc: noskill=1/1 (100%), prefill=100% (12/12)
- citation-check: noskill=1/1 (100%), prefill=92% (11/12)
- earthquake-phase-association: noskill=1/1 (100%), prefill=75% (9/12)
- econ-detrending-correlation: noskill=1/1 (100%), prefill=75% (9/12)

4.3 Tasks that fail regardless (both <30%)

- azure-bgp-oscillation-route-leak: prefill=20% (1/5), noskill=0/1 (0%)
- civ6-adjacency-optimizer: prefill=9% (1/11), noskill=0/1 (0%)
- enterprise-information-search: prefill=0% (0/12), noskill=0/1 (0%)

4.4 Retracted findings

The following findings from v2 are RETRACTED due to rate-limit contamination:

- 'citation-check: 0% pass rate with skills' - ALL 48 trials were rate-limited
- 'civ6-adjacency-optimizer: 0% pass rate' - ALL 48 trials were rate-limited
- 'azure-bgp: 0% pass rate' - ALL 48 trials were rate-limited
- 'GT-only paradox (14%)' - 11/14 gtonly trials were rate-limited
- 'Noise provides useful diversity' - based on fake GT-only comparison

5. Noise Mode (Clean Data)

Noise Mode	Pass/Total	Rate
random	30/49	61.2%
hard	35/50	70.0%
easy	31/47	66.0%

Caveat: With 11 clean tasks and N~38 per group, the 5pp spread (68-73%) is likely within noise. No statistically significant difference between noise modes.

6. Pool Size (Clean Data)

Pool Size	Pass/Total	Rate	Prompt
5	23/37	62.2%	17.6K
10	24/35	68.6%	21.4K
20	25/38	65.8%	32.7K
50	24/36	66.7%	57.9K

No clear pool size effect in clean data. Differences are small and likely noise.

7. Required Next Steps

This experiment is INCOMPLETE. The following must be done before any conclusions:

1. RERUN RATE-LIMITED TRIALS: 167 prefill + 13 baseline configs are garbage. Must rerun when rate limit resets, or use API key instead of Max Plan.
2. BASELINE $N \geq 5$: Current $N=1$ baselines are coin-flips. Need 5+ seeds per task for noskill and gtonly. Without this, no prefill-vs-baseline comparison is valid.
3. ATTENTION DILUTION CONTROL: Add a 'random-long-text' baseline - inject equal-length irrelevant text instead of noise skills. If it performs like prefill, then 'noise helps' is actually just 'dilution of bad GT instructions'.
4. ERROR TAXONOMY: Classify failures as TIMEOUT / FORMAT_MISMATCH / LOGIC_ERROR / PATH_ERROR / TOOL_MISUSE / RATE_LIMITED. Without this, mechanism analysis is impossible.
5. TASK-LEVEL PAIRED ANALYSIS: Compare per-task means, not config-level aggregates. Use paired bootstrap or McNemar test. The current aggregation inflates confidence.
6. STRATIFY BY TASK TYPE: Separate 'model already knows' (noskill=PASS) from 'model needs help' (noskill=FAIL). Skill injection likely works very differently for these two groups.
7. SKILL QUALITY AUDIT: Review GT SKILL.md files for the failing tasks. If the skill documents themselves are flawed, that explains low GT-only performance without needing exotic 'diversity' hypotheses.

8. Appendix: Data Contamination Summary

Task	Clean	Rate-Ltd	% Poison
3d-scan-calc	48	0	0%
adaptive-cruise-control	43	0	0%
azure-bgp-oscillation-route-le..	5	7	58%
citation-check	12	0	0%
civ6-adjacency-optimizer	11	1	8%
dapt-intrusion-detection	12	0	0%
dialogue-parser	12	0	0%
earthquake-phase-association	12	0	0%
earthquake-plate-calculation	12	0	0%
econ-detrending-correlation	12	0	0%
energy-ac-optimal-power-flow	5	7	58%
energy-market-pricing	5	7	58%
enterprise-information-search	12	0	0%
exceltable-in-ppt	12	0	0%